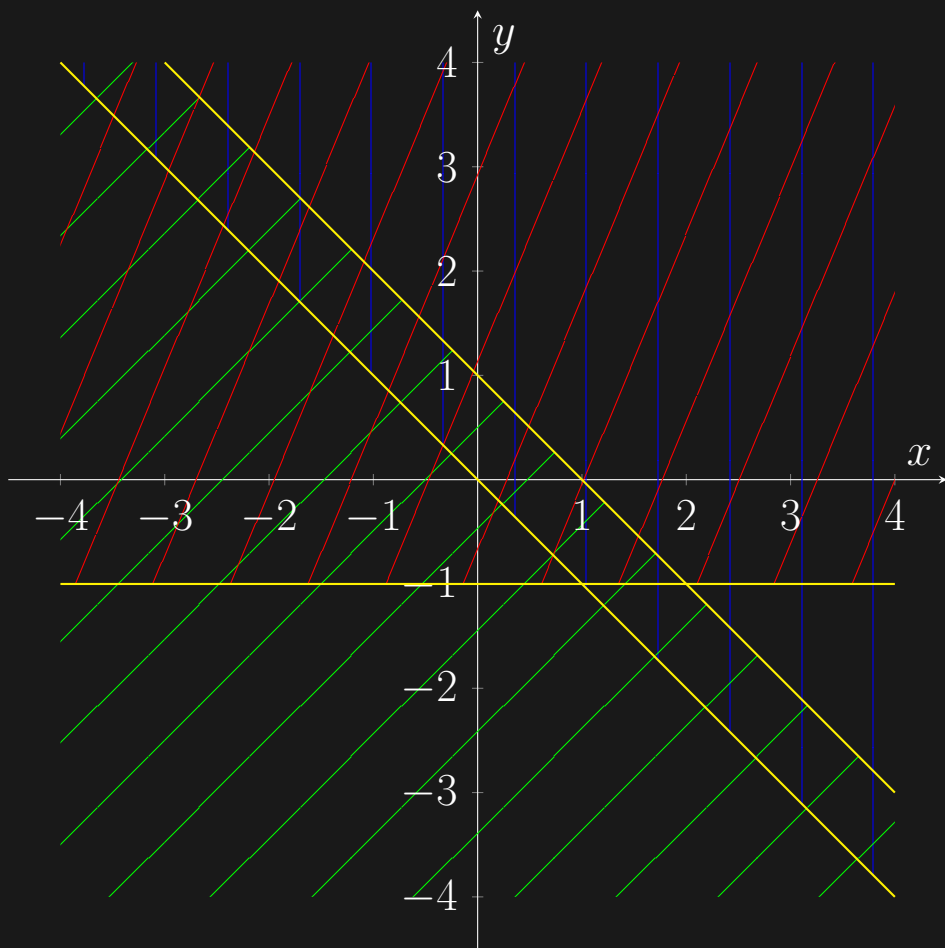


$$\iint_D \sqrt{x+y} \, dx \, dy \quad D = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x+y \leq 1, y \geq -1\}$$

$$\begin{cases} 0 \leq x+y \\ x+y \leq 1 \\ y \geq -1 \end{cases} \Leftrightarrow \begin{cases} -y \leq x \\ x \leq 1-y \\ y \geq -1 \end{cases} \Leftrightarrow \begin{cases} -y \leq x \leq 1-y \\ y \geq -1 \end{cases}$$



$$\begin{aligned} \int_{-1}^{\infty} \int_{-y}^{1-y} \sqrt{x+y} \, dx \, dy &= \int_{-1}^{\infty} \int_{-y}^{1-y} \sqrt{x+y} (x+y)' \, dx \, dy = \\ &= \int_{-1}^{\infty} \frac{(x+y)^{\frac{3}{2}}}{\frac{3}{2}} \Big|_{-y}^{1-y} dy = \int_{-1}^{\infty} \frac{2}{3} - \frac{0}{3} dy = \frac{2}{3} y \Big|_{-1}^{\infty} = \infty \end{aligned}$$

WIP